Horizontal launches

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Find flight time from vertical motion and then horizontal range.

SUCCESS CRITERIA - I can...

1. Separate vertical and horizontal givens.
2. Find or use fall time before calculating range.
3. Check units and interpret the landing point.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “horizontal” and “vertical” to explain your model or result. Sentence starters are optional.

horizontal: along a level line, usually the x direction

vertical: up and down, usually the y direction

displacement: change from initial position to final position, with direction

01 Bring back an earlier idea

A spherical planet has surface field 75 N/kg.
Find the field at center distance 5 R, where R is the planet radius.

Word help: radius = distance from the center to the edge of a circle or sphere

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $h = \frac{1}{2}gt^2$; range = $v_x t$.
Horizontal launch; h is the positive drop, t is fall time; g = 10 m/s². CPA may use the supplied time.

Worked reasoning: A horizontal launch falls 20 m with g=10 m/s². From $20=5\ t^2$, t=2 s. At horizontal speed 3 m/s, range=6 m.

Your turn: Draw a diagram labeling height and range. Number the two solution steps to show why time comes first.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

03 Look and explain

On the diagram, label height h, horizontal range and initial horizontal velocity. Which direction determines the fall time?

Sentence starter: I found … [unit]. This value means … in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Build the solution

A ball launches at 4 m/s horizontally. The supplied fall time from its platform is 2 s.
Calculate range. A second ball launches at 6 m/s from the same platform: find its range.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

Two platforms have heights 5 and 20 m. Supplied fall times are 1 and 2 s. At the same launch speed, does quadrupling height quadruple range?
Explain.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

A 6 kg cart starts from rest under 48 N net force right for 5 s.
Find acceleration and final velocity.
State one assumption.

Word help: net force = the vector sum of all forces on one object

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

07 Exit check

A horizontal launch at 5 m/s stays in air 3 s. Where does it land relative to the platform base?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A target is 10 m from a platform whose fall time is 2 s. **Choose** the horizontal launch speed, then predict where half that speed lands.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3